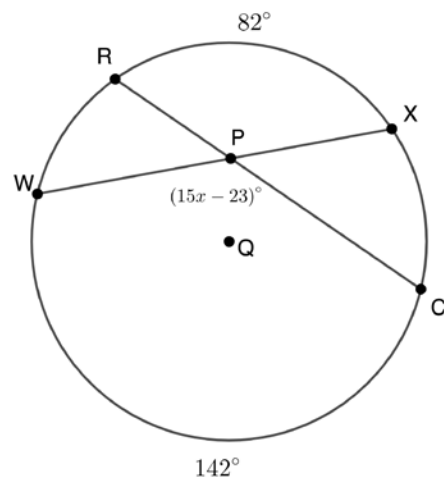
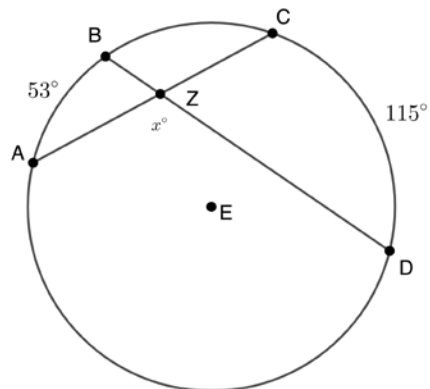


For questions 1-3, Circle Q is shown, where chords $\overline{RC}$ and $\overline{WX}$ intersect at point P , $m\widehat{CW} = 142^\circ$, $m\widehat{RX} = 82^\circ$, and $m\angle WPC = (15x - 23)^\circ$.

- Find the $m\angle WPC$.
- Write a formula to find the value of x .
- Find the value of x .

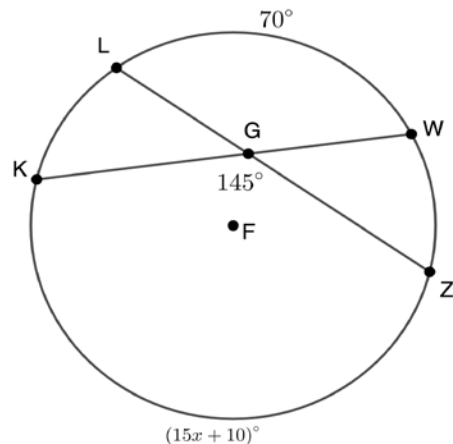


- Circle E is shown, where chords $\overline{BD}$ and $\overline{CA}$ intersect at point Z . Find the value of x for $\angle AZD$, given $m\widehat{CD} = 115^\circ$ and $m\widehat{AB} = 53^\circ$.



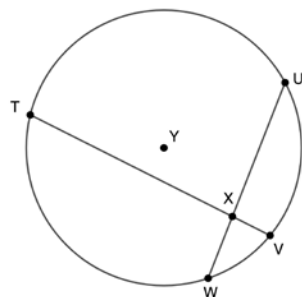
For questions 5-7, Circle F is shown, where chords $\overline{LZ}$ and $\overline{KW}$ intersect at point G , $m\widehat{LW} = 70^\circ$, $m\widehat{KZ} = (15x + 10)^\circ$, and $m\angle KGZ = 145^\circ$.

- Write a formula to find the value of x .
- Find the value of x .
- Find the $m\widehat{KZ}$.



For questions 8-10, Circle Y is shown, where chords $\overline{VT}$ and $\overline{UW}$ intersect at point X .

8. If $m\widehat{TU} = 128^\circ$ and $m\widehat{VW} = 30^\circ$, what is the measure of $\angle TXU$?



9. If $m\widehat{TW} = 152^\circ$ and $m\widehat{UV} = 40^\circ$, what is the measure of $\angle TXW$?

10. If $m\widehat{VW} = 56^\circ$ and $m\angle TXU = 78^\circ$, what is the measure of $\widehat{TU}$?